



Semester I Examinations 2021/22

Exam Code(s) 4BMS2, 4BS2, 4BPT2, 1HA1, 1AL1
Exam(s) 4th Science

Module(s) Algebraic Foundations of Quantum Computing
Module Code(s) MA4102

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Instructions Answer **FOUR** questions.

Duration 2 hours
No. of Pages 4 pages (including this cover page)
Discipline Mathematics

Requirements:

Release to Library	Yes ☒	No ☐
MCQ	Yes ☐	No ☒
Statistical/Log Tables	Yes ☒	No ☐

1. (i) Consider the **qutrit** state given by

$$|\psi\rangle = \frac{1}{3} (2|0\rangle - 2i|1\rangle + i|2\rangle).$$

- (a) What is the probability of measuring the particle in the $|2\rangle$ state?
 (b) Calculate the density matrix ρ_ψ .
 (c) If we operate on $|\psi\rangle$ using the unitary

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

calculate the subsequent probability of measuring $|2\rangle$.

[14 marks]

- (ii) Consider the **qubit** state

$$|\psi\rangle = \frac{1}{2} \left(\sqrt{3}|0\rangle + \frac{(1-i)}{\sqrt{2}}|1\rangle \right).$$

- (a) If $|\psi\rangle$ is represented on the Bloch Sphere, calculate the corresponding angles θ and ϕ .
 (b) If we measure using the orthonormal basis $|+\rangle$ and $|-\rangle$, calculate the probability of obtaining $|+\rangle$.

[11 marks]

2. Suppose we write a bipartite (2-qubit) state as a column vector with 4 components, using the standard basis $|00\rangle = (1, 0, 0, 0)^T$, etc. Calculate (a) the purity and (b) the entanglement (using the Von Neumann entropy of the reduced density matrix) for the second qubit in each of the following cases.

$$\text{(i)} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} \quad \text{(ii)} \quad \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ -1 \end{pmatrix} \quad \text{(iii)} \quad \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \\ \sqrt{-3} \end{pmatrix} \quad \text{(iv)} \quad \frac{1}{2\sqrt{2}} \begin{pmatrix} \sqrt{2} \\ \sqrt{3} \\ 1 \\ \sqrt{2} \end{pmatrix}$$

[25 marks]

3. (i) Given the orthonormal basis

$$|g\rangle = \frac{1}{\sqrt{7}}|0\rangle + i\frac{\sqrt{6}}{\sqrt{7}}|1\rangle, \quad |h\rangle = \frac{-\sqrt{6}}{\sqrt{7}}|0\rangle + i\frac{1}{\sqrt{7}}|1\rangle$$

calculate the associated Kraus operators M_g and M_h for the phase damping channel U given by

$$\begin{aligned} U(|10\rangle) &= |10\rangle & U(|11\rangle) &= |11\rangle \\ U(|00\rangle) &= \frac{1}{3}(2\sqrt{2}|00\rangle + i|01\rangle) \\ U(|01\rangle) &= \frac{1}{3}(i|00\rangle + 2\sqrt{2}|01\rangle) \end{aligned}$$

(where the second qubit represents the environment). [13 marks]

- (ii) For the positive operator-valued measure (POVM) given by

$$E_1 = \frac{1}{2}|0\rangle\langle 0|, \quad E_2 = \frac{1}{3}|+\rangle\langle +|, \quad E_3 = 1 - E_1 - E_2,$$

calculate the probability of each of the three outcomes when measuring the state

$$|\psi\rangle = \frac{-3}{5}|0\rangle + \frac{4}{5}|1\rangle.$$

[12 marks]

4. (i) Draw the quantum circuit corresponding to each of the following operations on the given input. (You do not need to calculate the resulting output.)

- (a) $(H \otimes H)CNOT(2, 1)(H \otimes I)|01\rangle$
- (b) $(I \otimes H^{\otimes 2})SWAP(1, 3)(CR_{\pi/3}(1, 2))|010\rangle$
- (c) $(X^{\otimes 2})(SWAP)(Y \otimes I)|11\rangle$

[15 marks]

- (ii) Calculate the Quantum Fourier Transform of the following qudit states:

$$(a) \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i \\ -i \end{pmatrix} \quad (b) \frac{1}{\sqrt{3}} \begin{pmatrix} i \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

[10 marks]

5. Suppose we are carrying out a Grover search of a list of length 32, for an item X which is guaranteed to occur exactly once.
- (i) Calculate the theoretical number of Grover iterates we should apply (this need not be a Natural Number). **[6 marks]**
 - (ii) Calculate how many (integer) iterations we should do to maximize our success probability, and calculate the success probability at that point. **[11 marks]**
 - (iii) How many iterations will give us the “second best” chance of finding X , and what is the associated probability? **[8 marks]**